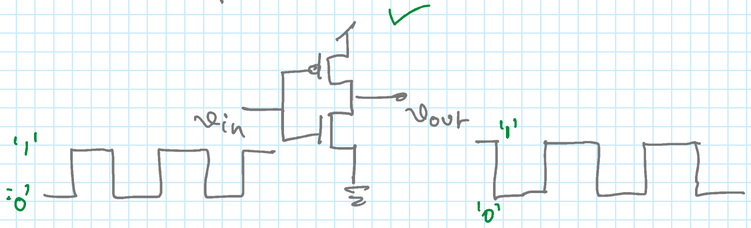
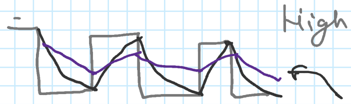
CML circuits design

CML → current mode logic → high speed digital circuits

CMOS logic → Inverter
logic gates → AND, NAND, NOR OR, XOR, XNOR
Dynamic logic → Latch, DFF

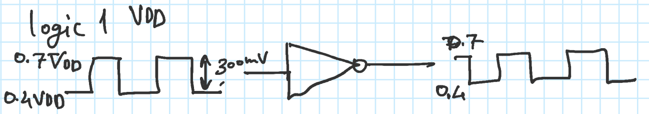
Inverter :





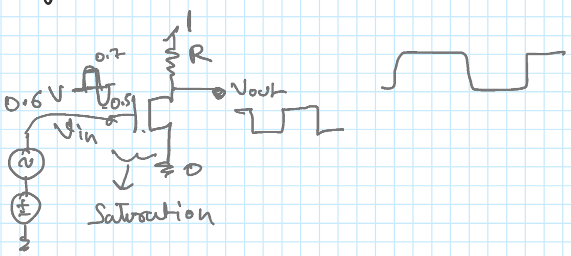
High speed circuits

TSMC 65nm → 3 Gb/s

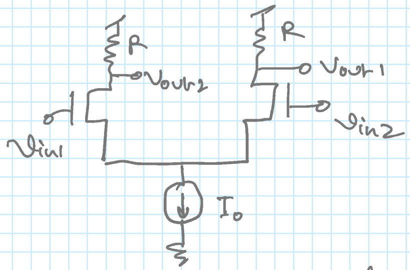


VDD − 1
300mV
→ 3–4 times
0

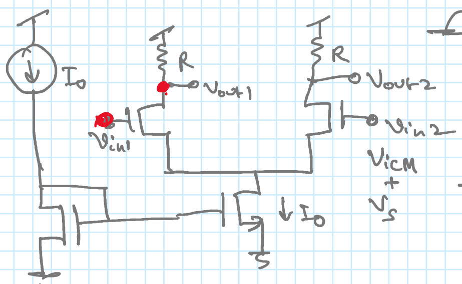
logic 0 → GND



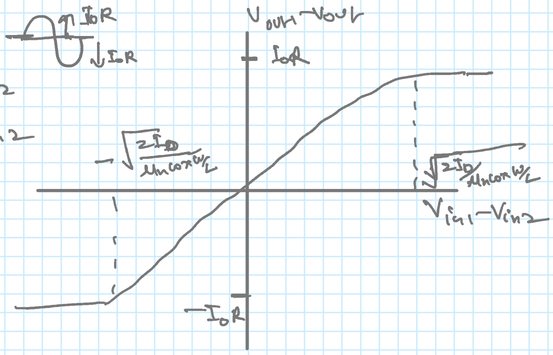
$v_{in1} - v_{in2} = A(V_{out1} - V_{out2})$



Differential amplifier

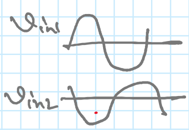


CML inverter



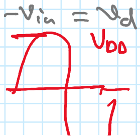
$v_{icm} = \frac{v_{in1} + v_{in2}}{2}$

$v_{in1,2} = v_{icm} \pm v_d/2$



$\Rightarrow v_{in1} - v_{in2} = v_d$

$v_{icm} + v_d/2 < V_{DD} - I_0R + V_T$

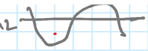


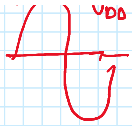
$V_{DD} < V_{DD} - I_0R + V_T$

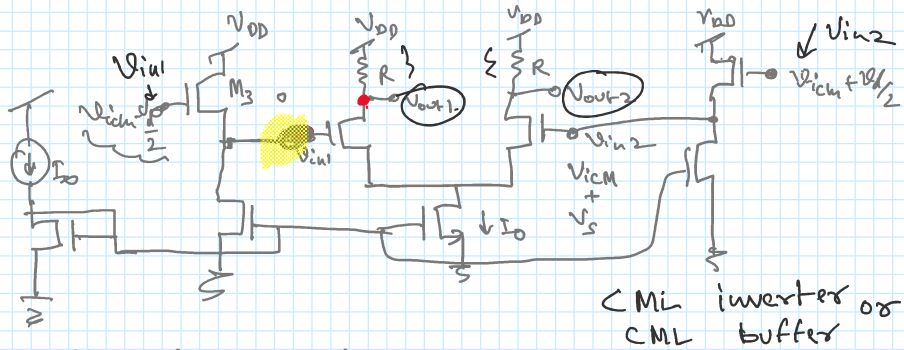
$\underline{v_{icm} + v_d/2} < \underline{V_{DD} - I_o R + V_T}$

$V_{DD} < V_{DD} - I_o R + V_T$

$\boxed{I_o R < V_T}$



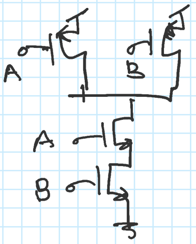




CML inverter or CML buffer

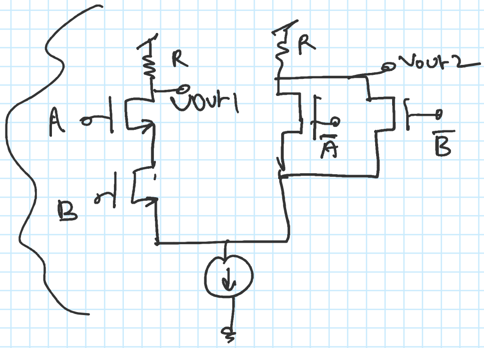
$\underline{\underline{V_{DD}}} \underline{- V_{Dsat3} - V_{T3}} < \underline{V_{DD} - I_o R + V_{T1}}$

$I_o R < V_{T1} + V_{T3} + V_{Dsat3}$



NAND

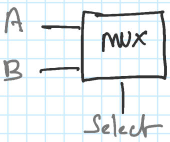
CML → PMOS → slower 2-3 times w.r.t NMOS



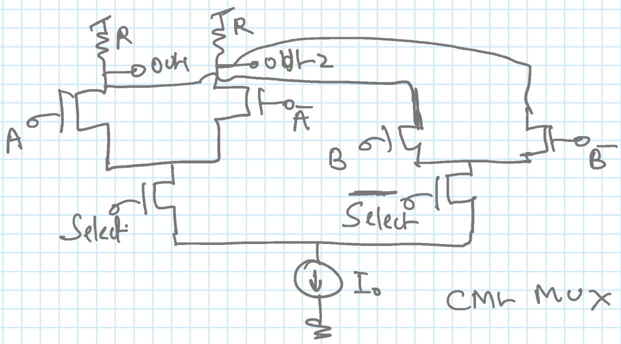
A. B =

NAND →
NOR →
OR →
AND →

CML MUX



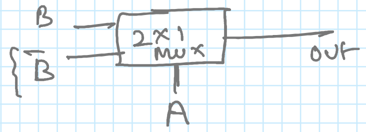
Select	output
0	A
1	B



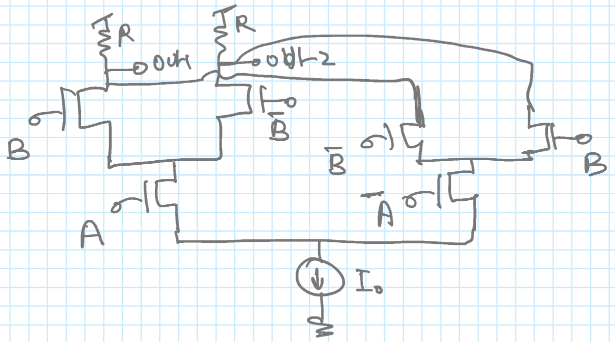
CML MUX

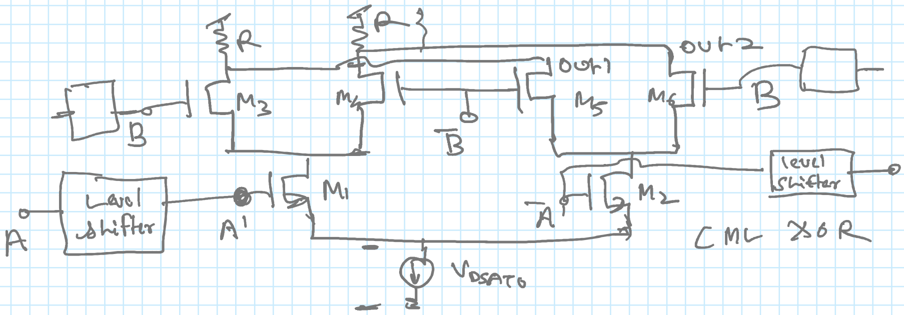
CML XOR

A	B		Output
0	0	B	0
0	1		1
1	0	$\bar{B}$	1
1	1		0



2x1 MUX an XOR

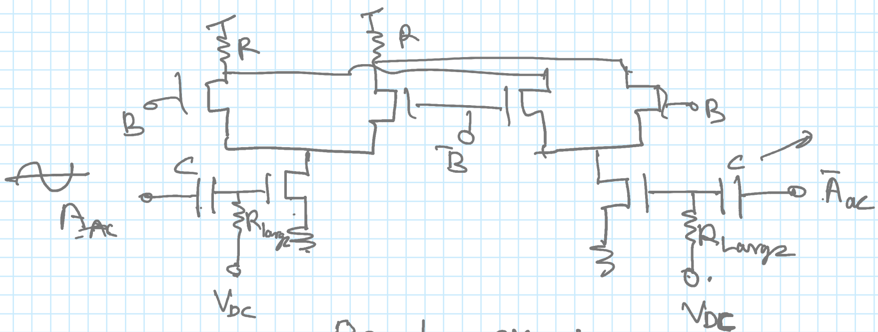




CML XOR

V_{ACM} : $V_{T1} + V_{DSAT1} + V_{DSAT0} < V_{ACM} < V_{BCM} - V_{DSAT3} - V_{T3} + V_{T1}$

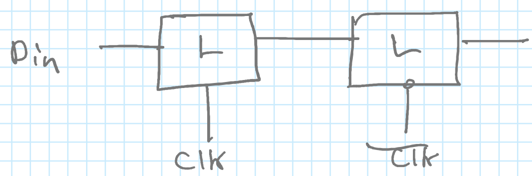
over head voltage problem: Many transistor in stack.

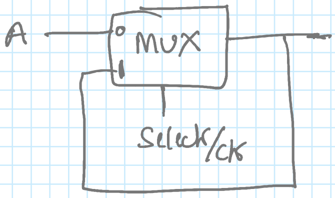


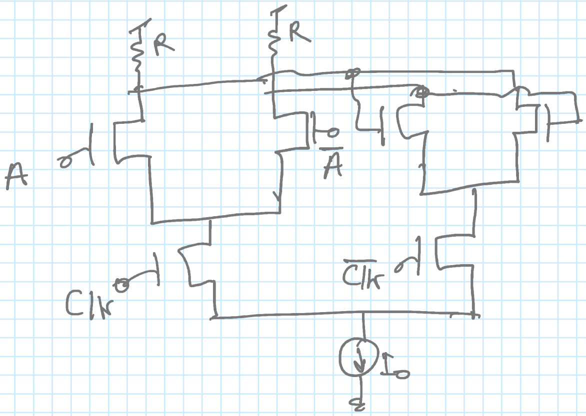
Pseudo-CML.

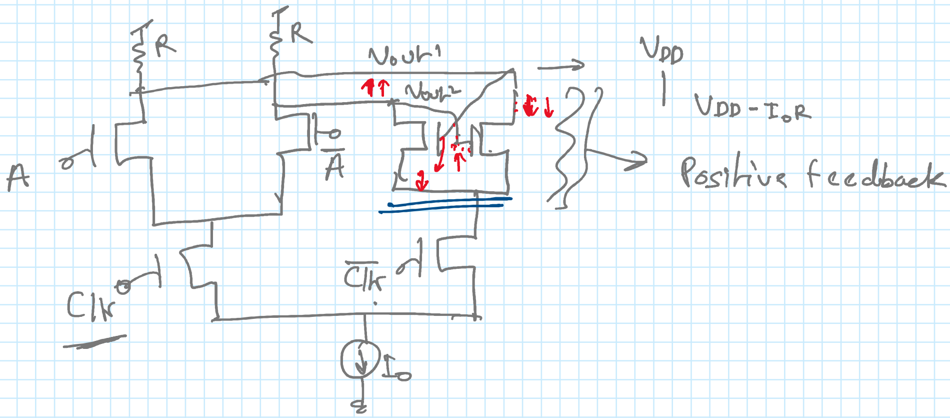
CML DFF

D-FF

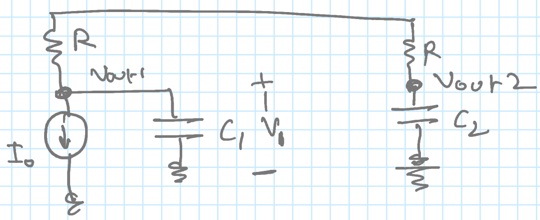








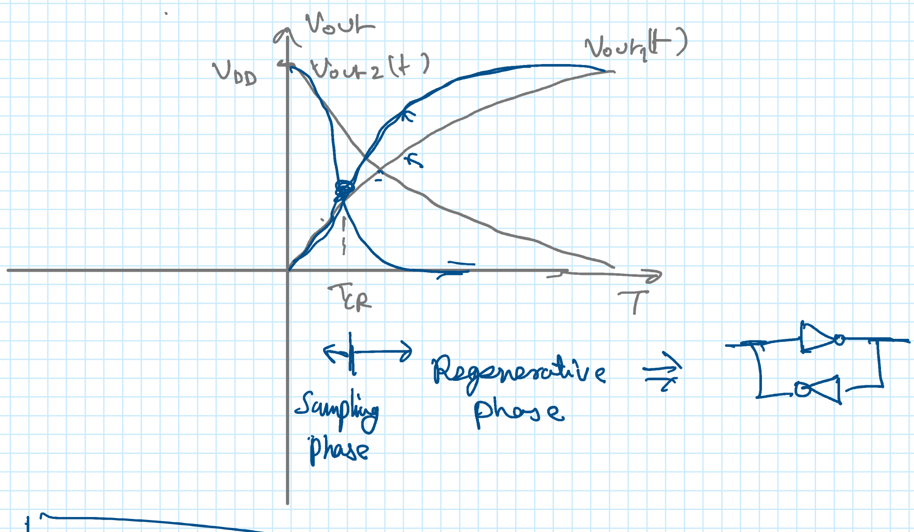
CML latch

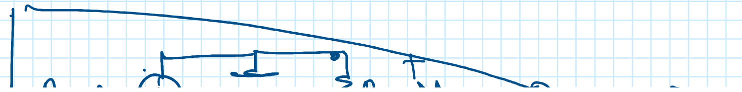


$$\frac{V_{DD} - V_1}{R} - C_1 \frac{dv_1}{dt} = I_o \quad , \; V_1(0) = V_{DD}$$

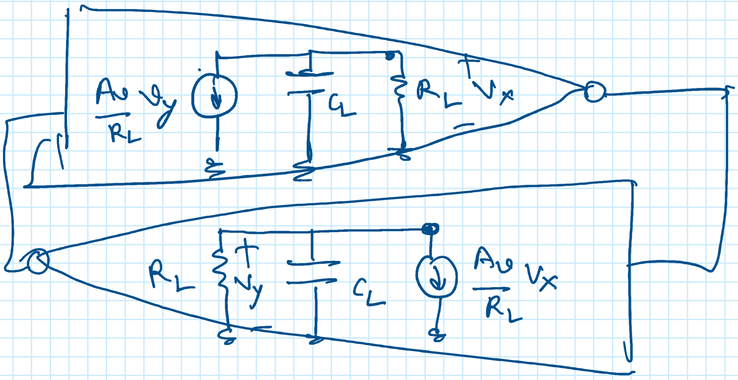
At Vout1 node

$$V_1(t) = V_{DD} - I_o R \left(1 - e^{-t/RC_1}\right)$$





A_v $v_y = -C_L \left(\frac{dv_x}{dt}\right) - v_x$



$$\frac{A_v}{R_L} v_y = -C_L\left(\frac{dv_x}{dt}\right) - \frac{v_x}{R} \quad \text{—(1)}$$

$$\frac{A_v}{R_L} v_x = -C_L \frac{dv_y}{dt} - \frac{v_y}{R} \quad \text{—(2)}$$

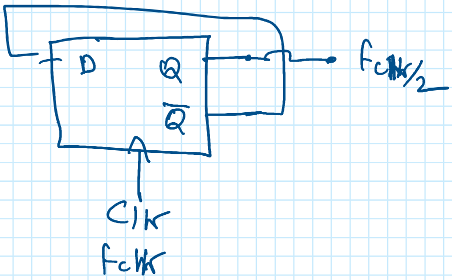
$$\Delta V = V_x - V_y$$

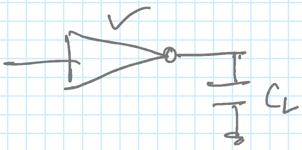
$$\Delta V = \Delta V_0 \, e^{\frac{(A_v-1)t}{\tau}} \qquad \tau = R_L C_L$$

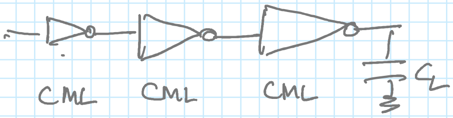
$$\tau_{latch} = \frac{\tau}{A_v - 1} = \frac{R_L C_L}{A_v} = \frac{C_L}{g_m}$$

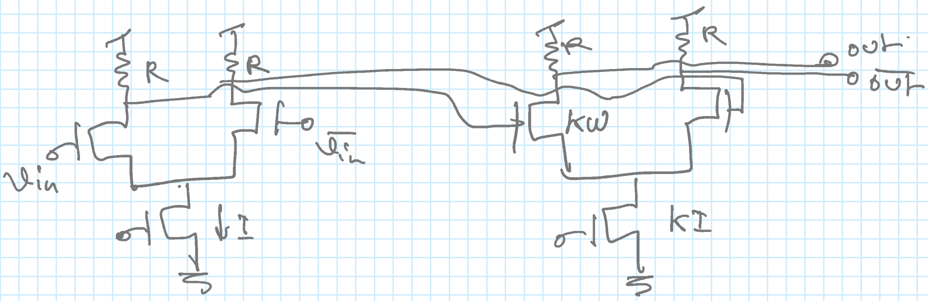
$$C_L = k_1 W L C_{ox} \qquad g_m = k_2 g_m = k_2 \mu_n C_{ox} \frac{W}{L} V_{eff}$$

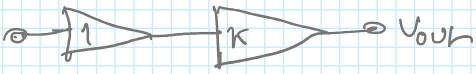
$$\boxed{T_{Latch} = \frac{C_L}{g_m} \ln\left(\frac{\Delta V_{logic}}{\Delta V_0}\right)} \rightarrow$$

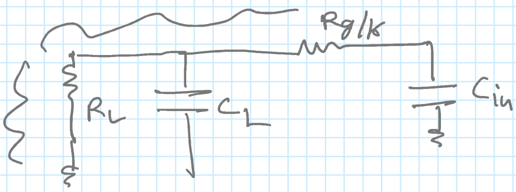












$$\tau = \tau_1 + \tau_2 = R_L(C_{gd} + C_{db}) + \left(R_L + R_g/_K\right)\left[K C_{gs} + (1+g_m R_L) K C_{gd}\right]$$

$A_v = -g_m R_L \qquad g_m = I/V_{eff} \qquad R_L = \frac{\Delta V}{I_T}$

$$A_v = -\frac{\Delta V}{V_{eff}}$$

$$\boxed{\tau_C = \Delta V \frac{C_{gd} + C_{db} + C_{int}}{I_T} + \left(K + \frac{R_g}{R_L}\right)\Delta V \frac{C_{gs} + (1 - A_v)C_{gd}}{I_T}}$$

$$\left\{ J = \frac{0.3\,mA}{\mu m} \right\} \rightarrow$$ Constant over technolog nodes for gate delay.

$$\Delta V_{min} = \left[V_{gs}\left(I_{DS} = 0.15\,mA/\mu m\right) - V_T\right] \longrightarrow$$ Scale with L